



Divisor Detective

CPU TIME LIMIT

1 second

MEMORY LIMIT

1024 MB

DIFFICULTY

medium (2.8)

Ådne is a software engineer who loves solving algorithmic puzzles. His latest project is a tool called "Divisor Detective" that analyzes number patterns.

While testing the tool with his colleague, they encountered an interesting mathematical challenge:

"For each integer from 1 to N , we need to count how many divisors it has, then sum up all those counts. But with these large inputs, we need an efficient algorithm to compute this quickly!"

Can you help them determine the sum of the divisor counts for all integers from 1 to N ?

Input

The input consists of a single line containing one integer N .

Output

Output a single integer: the sum of the number of positive divisors of every integer from 1 up to N .

Limits

- $1 \leq N \leq 10^{10}$



Sample Input 1

6

Sample Output 1

14

Sample Input 2

10

Sample Output 2

27
